Lecture Flow

Program: Diploma in Cyber Security

Courses: CompTia A+

CompTia A+ Lecture Flow:

1. Hardware and Components Basics Understanding of Hardware and its components (Duration: 10 Hours)

1 .1 Day 1: Overview of Hardware ComponentsTheory (1 hour):Introduction to motherboards, CPUs, RAM, GPUs, storage (HDD, SSD).Importance of each component in a computer system.Assignment: Create a diagram of a PC with labeled hardware components.

1 .2 Day 2: Motherboard and CPUTheory (30 min):Types of motherboards and CPU socket compatibility.Functions of the CPU and chipset.Practical (30 min):Identify the motherboard slots and ports.

1 .3 Day 3: RAM and StorageTheory (30 min):Types of RAM (DDR3, DDR4, DDR5) and storage (HDD vs. SSD, NVMe).Practical (30 min):Remove and reinstall RAM and storage drives.

1 .4 Day 4: GPUs and Cooling SystemsTheory (30 min):Basics of GPUs and cooling systems (air vs. liquid cooling).Practical (30 min):Identify a GPU and cooling system in a PC setup.

1 .5 Day 5: Assemble and Disassemble a PCPractical (1 hour):Full hands-on session: Assemble and disassemble a desktop PC.Task: Document the process with labeled steps.

2. A+ - Peripherals and Power Supply (Duration: 10 Hours)

2 .1 Day 1: Overview of PeripheralsTheory (1 hour):Functions of printers, scanners, webcams, and their connection interfaces.

2 .2 Day 2: Power Supply SystemsTheory (30 min):Basics of SMPS and UPS systems.Practical (30 min):Identify power cables and connectors.

2 .3 Day 3: Peripheral TroubleshootingTheory (30 min):Common issues and solutions for peripherals.Practical (30 min):Troubleshoot a non-working printer.

2 .4 Day 4: Connecting and Testing PeripheralsPractical (1 hour):Set up and test peripherals like printers and webcams.Task: Resolve connectivity issues for a given setup.

2 .5 Operating System information, windows, Linux, macOS installation and adjust OS settings

3. A+ - Understanding and maintenance of Networks (Duration: 5 Hours)

3 .1 Day 1: Introduction to Networking HardwareTheory (1 hour):Overview of modems, routers, switches, access points, and network cables.

3 .2 Day 2: Routers and SwitchesTheory (30 min):Basic functions and differences between routers and switches.Practical (30 min):Identify ports and LEDs on a router and switch.

3 .3 Day 3: Configuring Basic Network HardwareTheory (30 min):IP addressing basics and network configuration.Practical (30 min):Set up a router with default settings.

3 .4 Day 4: Network Cable Types and ToolsTheory (30 min):Types of cables (Ethernet, fiber optics) and tools (RJ45 crimper).Practical (30 min):Create a straight-through Ethernet cable

3 .5 Day 5: Network Testing Practical (1 hour): Connect devices to a network and test connectivity (ping, traceroute). Task: Write a report on how to set up and test a basic network.

4. A+ - Troubleshooting and Helpdesk (Duration: 5 Hours)

4 .1 Day 1: Tools for TroubleshootingTheory (1 hour):Overview of tools (multimeter, POST card, software diagnostics).

4 .2 Day 2: Diagnosing Hardware FailuresTheory (30 min):Common hardware issues and their symptoms.Practical (30 min):Test components using a multimeter.

4 .3 Day 3: BIOS and POST Troubleshooting Theory (30 min): Understanding BIOS and POST errors. Practical (30 min): Analyze POST error codes on a given setup.

4 .4 Day 4: Component-Specific Troubleshooting Practical (1 hour): Identify and fix issues in faulty components (RAM, GPU, storage). Task: Create a checklist for diagnosing hardware issues.

4 .5 Day 5: Case Study Practical (1 hour): Work on a scenario involving multiple hardware issues.